EE 3072.1/372.1 **Student Name:** _____
Dynamic Systems and Control
Midterm Exam Solutions **Student Number:** _____
April 12, 2021
Time Limit: 75 Minutes

Questions

1. (20 points) Consider the linear system given by the following differential equation.

$$y^{(4)} + 3y^{(3)} + 2\ddot{y} + 3\dot{y} + 2y = \dot{u} - u,$$

where u is the input and y the output.

Is the system stable? Show your work to justify your claim.

Note: $y^{(3)}$ and $y^{(4)}$ denote the third and the fourth derivatives of y, respectively.

Answer:

$\mathcal{L}\{y^{(4)} + 3y^{(3)} + 2\ddot{y} + 3\dot{y} + 2y\} = \mathcal{L}\{\dot{u} - u\}$

$(s^4 + 3s^3 + 2s^2 + 3s + 2)Y(s) = (s-1)U(s)$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{s-1}{s^4 + 3s^3 + 2s^2 + 3s + 2}$$

s^4	1	2	2
s^3	3	3	
s^2	1	2	
s^1	-3		
s^0	2		

Table 1: Routh-Hurwitz Table

Two sign changes in the first column $\implies$ The system is unstable

2. (20 points) Consider the unity *negative* feedback loop with plant $G(s) = \dfrac{1}{s^2+6s+8}$ and a gain K. Assume that all initial conditions are equal to zero.

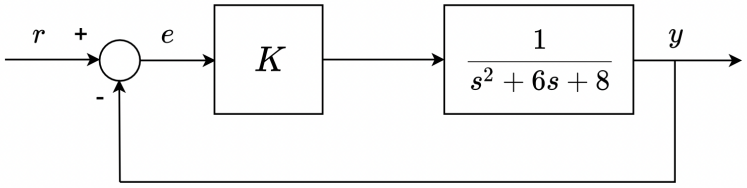


Figure 2: The feedback control system under consideration in Question 2.

a) (5 pts) What is the transfer function from r to y?

b) (5 pts) For what range of values of K is this closed system stable?

c) (5 pts) Suppose that we want the closed loop system to be underdamped with peak time $t_p < 2\pi$ seconds. Sketch the possible pole locations in the complex plane. Mark indicative regions or points if they are helpful.

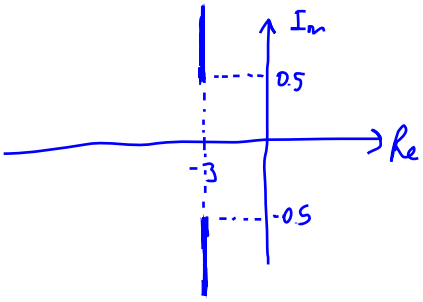
d) (5 pts) For what range of values of K is the closed loop system underdamped with peak time $t_p < 2\pi$ seconds?

Answer:

a) $G_{cl}(s) = \dfrac{K\frac{1}{s^2+6s+8}}{1+K\frac{1}{s^2+6s+8}} = \dfrac{K}{s^2+6s+8+K}$

b) Characteristic equation: $s^2+6s+8+K=0$. Necessary and sufficient condition for stability of a second order system: all coefficients must have the same sign: $8+K>0 \implies K>-8$.

c) $t_p = \frac{\pi}{\omega_d} \implies \omega_d > 0.5$. Closed loop poles are $s_{1,2} = \frac{-6\pm\sqrt{36-4(8+K)}}{2} = -3 \pm \sqrt{1-K}$. For an underdamped system, we require $1-K<0$ and closed loop poles are $-3 \pm j\sqrt{K-1}$.



d) In order to have $\omega_d > 0.5$, we require $\sqrt{K-1} > 1/2 \implies K > 5/4$.

3. (20 points) Suppose that you are given the single-cart system shown below.

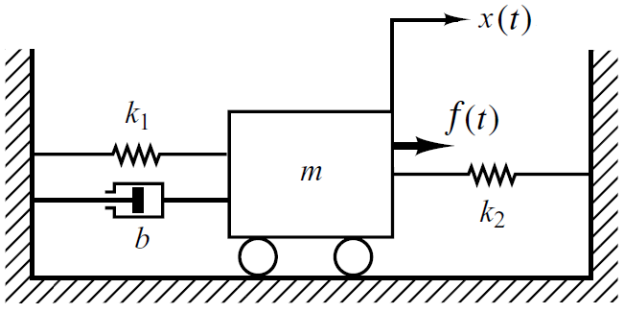


Figure 3: A mechanical system consisting of two springs and a dashpot.

Assume that there are zero initial conditions (i.e., the cart is initially at $x = 0$ and is not moving) and the cart is on a frictionless surface.

a) (6 pts) Find the ordinary differential equation that models the single-cart system shown.

b) (7 pts) Find the transfer function from f to x.

c) (7 pts) Let $f(t)$ be a unit step. What is the final value of $x(t)$? (i.e., find $x(\infty)$, the final position of the cart)

Answer:

a) $f = m\ddot{x} + k_1 x + k_2 x + b\dot{x}$.

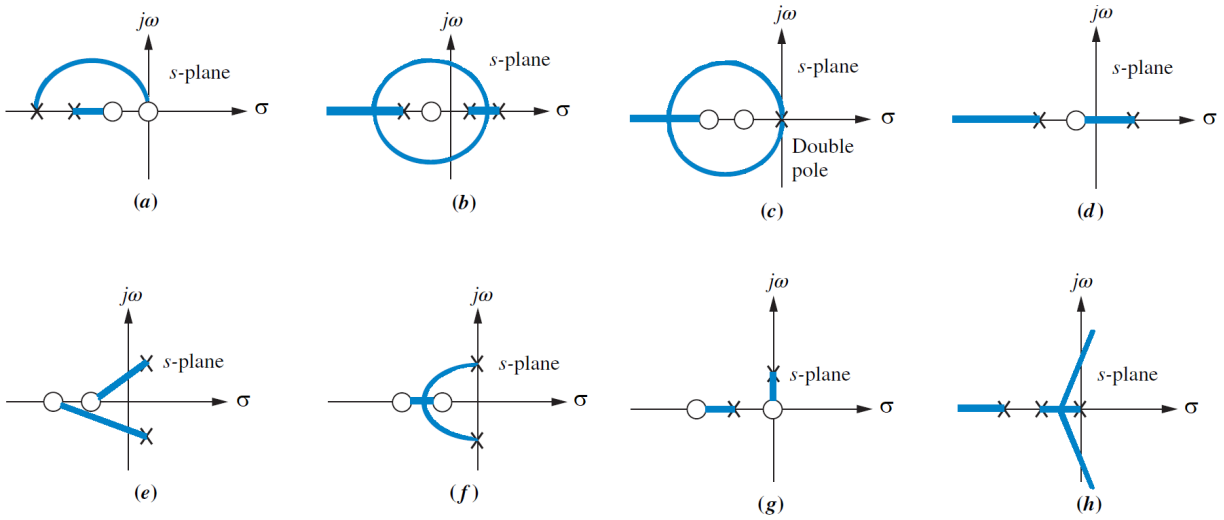
b) $F(s) = (ms^2 + bs + k_1 + k_2)X(s) \implies G(s) = \frac{X(s)}{F(s)} = \dfrac{1}{ms^2 + bs + k_1 + k_2}$

c) $f(t)$ is unit step, hence $F(s) = 1/s$

$$X(s) = G(s)F(s) = \frac{1}{s(ms^2 + bs + k_1 + k_2)}$$

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} sX(s) = \lim_{s\to 0} s\frac{1}{s(ms^2 + bs + k_1 + k_2)} = \frac{1}{k_1 + k_2}$$

4. (16 points) For each of the root loci shown in the figure below, state whether or not the sketch can be a root locus. If the sketch cannot be a root locus, explain why. Give all reasons.



Answer:

a) It cannot be. It is not symmetric about the real axis and root-locus on the real axis is incorrect.

b) It cannot be. Root locus should start at open loop poles and end at open loop zeros. Root-locus on the real axis is incorrect. There should be two asymptotes with angles 90° and 270°.

c) It cannot be. Root locus should start at open loop poles and end at open loop zeros. Root-locus on the real axis is incorrect.

d) It is a root locus.

e) It cannot be. It is not symmetric about the real axis and root-locus on the real axis is incorrect.

f) It is a root locus.

g) It cannot be. It is not symmetric about the real axis and root-locus on the real axis is incorrect.

h) It is a root locus.

5. (24 points) Consider the system shown in Figure 5a. The damping ratio of this system is 0.158 and the undamped natural frequency is 3.16 rad/sec. To improve the relative stability, we employ tachometer feedback. Figure 5b shows such a tachometer-feedback system.

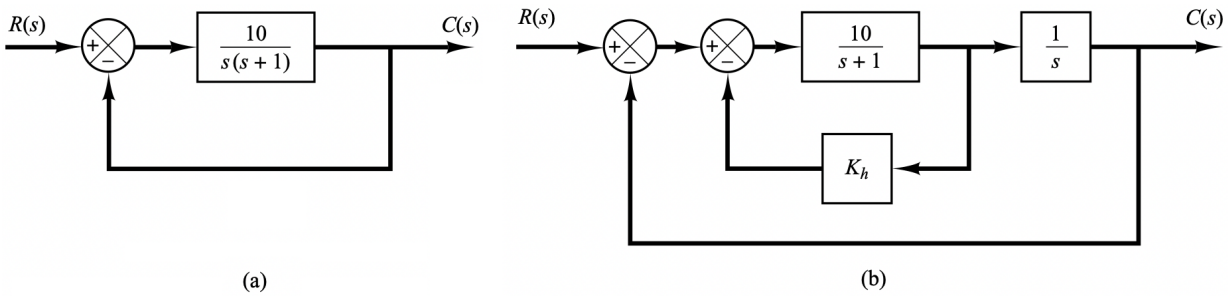


Figure 5: (a) Control system; (b) control system with tachometer feedback.

a) (4 pts) Determine the value of K_h so that the damping ratio of the system is 0.5.

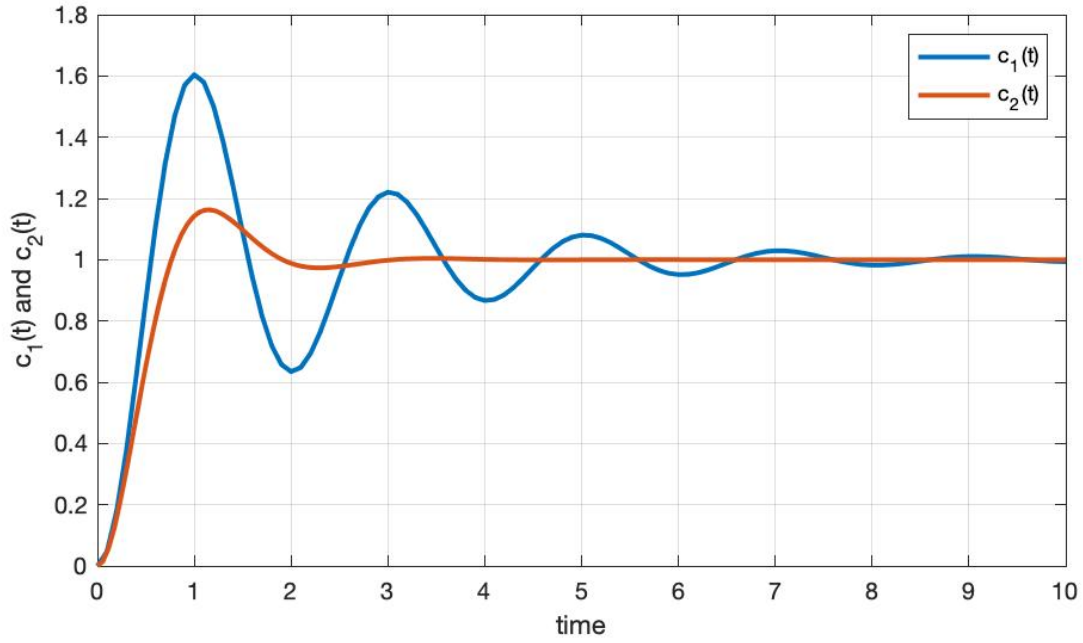
b) (8 pts) For both the original and tachometer-feedback systems, determine the peak time, rise time, overshoot and settling time (2% error band).

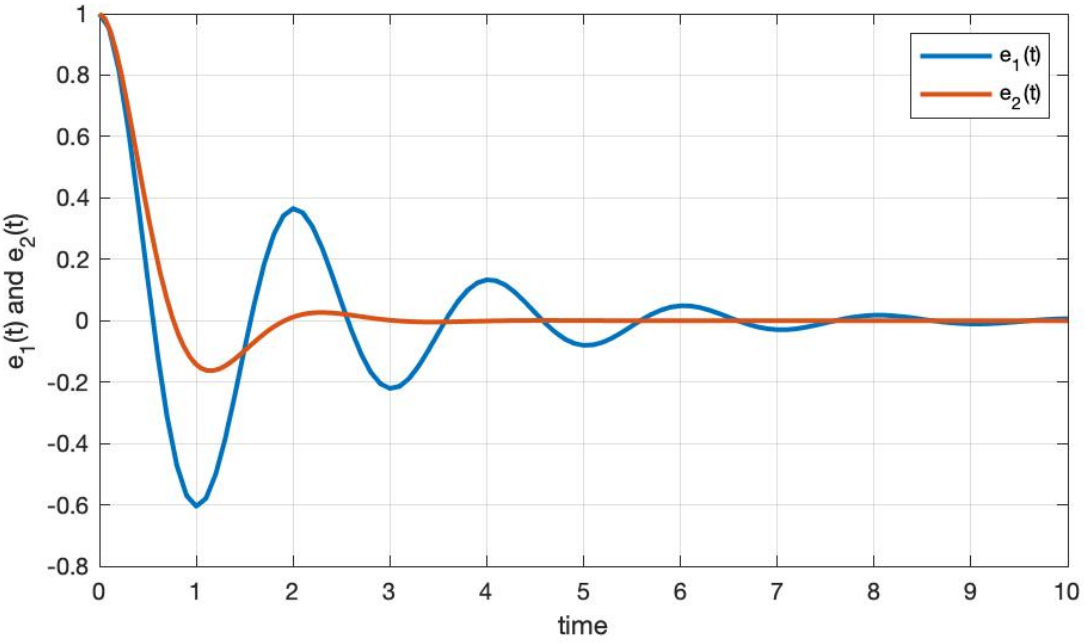
c) (12 pts) By using the values computed in the previous part, draw the *output-versus-time* and *error-versus-time* curves for both systems when the input is unit step.

Answer:

a) For the second system $G_{cl}(s) = \dfrac{\frac{10}{s+1+10K_h}\frac{1}{s}}{1+\frac{10}{s+1+10K_h}\frac{1}{s}} = \dfrac{10}{s^2+(1+10K_h)s+10}$. Since $2\zeta\omega_n = 1+10K_h$, we have $K_h = 0.216$.

b) For the first system $G_{cl1}(s) = \frac{10}{s^2+s+10}$ which implies $\omega_n = 3.16$, $\zeta = 0.158$, $\omega_d = \omega_n\sqrt{1-\zeta^2} = 3.120$, $\theta = \cos^{-1}\zeta = 1.41$, $t_r = (\pi-\theta)/\omega_d = 0.5543$, $t_p = \pi/\omega_d = 1$, $M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}100\% = 60\%$, $t_s = 4/\zeta\omega_n = 8.01$.
For the second system $G_{cl2}(s) = \frac{10}{s^2+(1+10K_h)s+10} = \frac{10}{s^2+3.16s+10}$ which implies $\omega_n = 3.16$, $\zeta = 0.5$, $\omega_d = \omega_n\sqrt{1-\zeta^2} = 2.74$, $\theta = \cos^{-1}\zeta = 1.05$, $t_r = (\pi-\theta)/\omega_d = 0.7653$, $t_p = \pi/\omega_d = 1.148$, $M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}100\% = 16.3\%$, $t_s = 4/\zeta\omega_n = 2.532$.


1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0 1 2 3 4 5 6 7 8 9 10
time
c1(t) and c2(t)
c1(t)
c2(t)


1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
0 1 2 3 4 5 6 7 8 9 10
time
e1(t) and e2(t)
e1(t)
e2(t)